```
> restart;#Time of interaction
```

```
> -ec^2/(6·Pi·ϵ·c^3)·(ec^2/(4·Pi·ϵ·m·r^2))^2; #The Larmour's formula
```

$$-\frac{ec^6}{96\,\pi^3\,\epsilon^3\,c^3\,m^2\,r^4} \tag{1}$$

```
> ec := 1.602176634·10^-19; ϵ := 8.8541878128 × 10^-12;  m := 9.109383560 × 10^-31;
  rb := 5.292410932 × 10^-11; c := 299792458; mn := m·  1838.683; mp :=
  1.672621923·10^-27; mτ := 0.0189/c^2
```

$$ec := 1.602176634 \times 10^{-19}$$
$$\epsilon := 8.854187813 \times 10^{-12}$$
$$m := 9.109383560 \times 10^{-31}$$
$$rb := 5.292410932 \times 10^{-11}$$
$$c := 299792458$$
$$mn := 1.674926869 \times 10^{-27}$$
$$mp := 1.672621923 \times 10^{-27}$$
$$m\tau := 2.102908606 \times 10^{-19} \tag{2}$$

```
> E := ec^2/(8·Pi·ϵ·rb); EeV := E/ec ;
```

$$E := 2.179609239 \times 10^{-18}$$
$$EeV := 13.60405084 \tag{3}$$

```
> V := (ec^2/(4·Pi·ϵ·rb·m))^0.5; evalf(c/137);
```

$$V := 2.187559244 \times 10^{6}$$
$$2.188266117 \times 10^{6} \tag{4}$$

```
> T0 := -3·c^3·(4·Pi·ϵ·m/ec^2)^2·int(r^2, r = 0..rb); -3·c^3·rb^3/3·(4·Pi·ϵ·m/ec^2)^2;
```

$$T0 := -6.226964259 \times 10^{-11}$$
$$-6.226964257 \times 10^{-11} \tag{5}$$

```
> int(r^2, r = 0..1);
```

$$\frac{1}{3} \tag{6}$$

```
> T0 := (4·Pi· ϵ·m/ec^2)^2·(rb·c)^3
```

$$T0 := 6.226964257 \times 10^{-11} \tag{7}$$

```
> EeV/0.189; EeV/0.924; T := EeV/0.189·t0; T1 := EeV/0.924·t0;
  #There are 72 (15) quants rest mass (kinetic energy) of τ neutrino
  's energy that fully charges an Hydrogen's electron with energy
  in 4.5 (0.9) ns
```

$$71.97910497$$
$$14.72299874$$
$$T := 71.97910497\ t0$$
$$T1 := 14.72299874\ t0 \tag{8}$$

```
> (0.924/(0.0018 + 0.00038))^-1;
```

#Probably less than 3-5% of the fully charges an Hydrogen's electron energy is supplied by kinetic energy & rest mass of the graviton because there are more colision in the unit of time than τ neutrino (not 0.2%).

$$0.002359307359 \quad (9)$$

```
> mp·c^2/ec + 0.189 + 0.924 -(mn·c^2/ec + 0.0087 -0.0018-0.00038 + 0.007 + 0.017);
  #-1.293MeV
```

$$-1.2929787 \times 10^{6} \quad (10)$$

```
> -(mn·c^2/ec + 0.0087 -0.0018-0.00038 + 0.007 + 0.017) + mp·c^2/ec + m·c^2/ec + 0.007
  + 0.017;#-781.98KeV
```

$$-781980.8580 \quad (11)$$

```
>
>
```